

Customer Shopping Behavior Analysis

1. Project Overview

Objective: Analyze customer shopping behavior using Python, SQL and Power BI to identify purchasing patterns, customer preferences, spending habits, and business opportunities.

Tools Used:

- Python
- Pandas
- SQL (PostgreSQL)
- Power BI
- Jupyter Notebook

2. Dataset Summary

Dataset: Customer Shopping Behavior Dataset

Records: Approximately 3900

Columns: 18

Key fields include Customer ID, Age, Gender, Item Purchased, Category, Purchase Amount, Location, Season, Review Rating, Payment Method, Subscription Status, and Frequency of Purchases.

3. EDA using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used `data.info()` to check structure and `data.describe()` for summary statistics.

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.
- **Column Standardization:** Renamed columns to snake case for better readability and documentation.
- **Feature Engineering:**
 - Created `age_group` column by binning customer ages.
 - Created `purchase_frequency_days` column from purchase data.
- **Data Consistency Check:** Verified if `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used`.
- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL

We performed structured analysis in PostgreSQL to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
Total rows: 839 of 839 Query completed		

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

	item_purchased text	average_product_rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

	subscription_status text	total_customer bigint	average_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased text	count bigint	sum bigint	discount_rate bigint
1	Hat	154	77	50
2	Sneakers	145	72	49
3	Coat	161	79	49
4	Sweater	164	79	48
5	Pants	171	81	47

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment text	Number of customer bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. Top 3 Products per Category – Listed the most purchased products within each category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. Repeat Buyers & Subscriptions – Checked whether customers with >5 purchases are more likely to subscribe.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. Revenue by Age Group – Calculated total revenue contribution of each age group.

	age_group text	revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5. Dashboard using Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.



6. Business Recommendations

1. **Maintain Inventory :** Keep sufficient stock of high-selling product to avoid shortages and increase sales.
2. **Seasonal Promotions:** Launch discount and marketing campaigns during peak shopping season to boost revenue.
3. **Customer Loyalty Programs:** Offer rewards, discounts and exclusive benefits to encourage repeat purchases.
4. **Personalized Recommendations:** suggest products based on customers previous purchases to improve sales and customer experience.
5. **Optimize Payment Option:** Promote popular payment methods and ensure a smooth checkout process.
6. **Targeted Marketing:** Focus promotions efforts on high-performing locations and customer segments for better results.